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JavaScript
Lecture#1st

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What is JavaScript?

- It is an interpreted, full-fledged programming language that enables dynamic interactivity on websites when applied to an HTML document.
- JavaScript is the most popular and widely used client-side scripting language.
- JavaScript is designed to add interactivity and dynamic effects to the web pages by manipulating the content returned from a web server.
- JavaScript is officially maintained by ECMA (European Computer Manufacturers Association) as ECMAScript. ECMAScript 6 (or ES6) is the latest major version of the ECMAScript standard

History of JavaScript

- Brendan Eich, a Netscape Communications Corporation programmer, created JavaScript in September 1995. It took Eich only 10 days to develop the scripting language, then known as Mocha.
- Later, the marketing team replaced the name with 'LiveScript'. But, due to trademark reasons and certain other reasons, in December 1995, the language was finally renamed to 'JavaScript'. From then, JavaScript came into existence.

What can in-browser JavaScript do?

- In-browser JavaScript can do everything related to webpage manipulation, interaction with the user, and the webserver.
- Add new HTML element to the page, change the existing content, modify styles.
- React to user actions, run on mouse clicks, pointer movements, key presses.
- Send requests over the network to remote servers, download and upload files
- Get and set cookies, ask questions to the visitor, show messages.
- Remember the data on the client-side (“local storage”).

What CAN’T in-browser JavaScript do?

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- JavaScript’s abilities in the browser are limited for the sake of the user’s safety. The aim is to prevent an evil webpage from accessing private information or harming the user’s data.
- JavaScript on a webpage may not read/write arbitrary files on the hard disk, copy them or execute programs. It has no direct access to OS system functions.
- Different tabs/windows generally do not know about each other, JavaScript from one page may not access the other if they come from different sites
- JavaScript can easily communicate over the net to the server but it requires explicit agreement from the remote side. Once again, that’s a safety limitation.

Features of JavaScript

- All popular web browsers support JavaScript as they provide built-in execution environments.
- JavaScript is a weakly typed language, where certain types are implicitly cast (depending on the operation).
- JavaScript is an object-oriented programming language.
- It is a light-weighted and interpreted language.
- It is a case-sensitive language.
- It provides good control to the users over the web browsers.

Variable

- A JavaScript variable is simply a name of storage location

```
<script>
var x = 10;
var y = 20.2;
var z=“Ali”;
document.write(z);
</script>
```

Naming Conventions for JavaScript Variables

- A variable name must start with a letter, underscore (_).
- A variable name cannot start with a number.
- A variable name can only contain alpha-numeric characters (A-z, 0-9) and underscores.
- A variable name cannot contain spaces.
- A variable name cannot be a JavaScript keyword or a JavaScript reserved word

The let and const Keywords

- The const keyword works exactly the same as let, except that variables declared using const keyword cannot be reassigned later in the code. Here's an example:

```
Example
let name = “Riaz Ahmad";
let age = 30;
let isStudent = true;
// Declaring constant
const PI = 3.14;
console.log(PI); // 3.14
// Trying to reassign
PI = 10; // error
```

Var, constant, and let Key

DEMO-1

Conditional Statements

- In coding, we want to perform different actions for different decisions. For that we can use conditional statements in our code.
- In JavaScript we have the following conditional statements:
  - If is used to specify a block of code to be executed, if a specified condition is true
  - else is used to specify a block of code to be executed, if the same condition is false
  - if else if is used to specify a new condition to test, if the first condition is false
  - switch is used to specify many alternative blocks of code to be executed

Conditional Statements

DEMO-2

JavaScript Array

- **JavaScript array** is an object that represents a collection of similar type of elements.
- There are 3 ways to construct array in JavaScript
  - By array literal
  - By creating instance of Array directly (using new keyword)
  - By using an Array constructor (using new keyword)

JavaScript array literal

- The syntax of creating array using array literal is given below:
- var arrayname=[value1,value2.....valueN];
- As you can see, values are contained inside [ ] and separated by , (comma).
- Let's see the simple example of creating and using array in JavaScript.

```
<script>
var emp=[“Ali",“Ahmad",“Kareem"];
document.write(emp[0] + "<br/>");
document.write(emp[1] + "<br/>");
}
</script>
```

JavaScript Array directly (new keyword)

- The syntax of creating array directly is given below:
- var arrayname=new Array();
- Here, **new keyword** is used to create instance of array.
- Let's see the example of creating array directly.

```
<script>
var emp = new Array();
emp[0] = "Saleem";
emp[1] = "Zabi";
emp[2] = "Raihan";
document.write(emp[0] + "<br>");
document.write(emp[1] + "<br>");
document.write(emp[2] + "<br>");
</script>
```

JavaScript array constructor (new keyword)

- Here, you need to create instance of array by passing arguments in constructor so that we don't have to provide value explicitly.
- The example of creating object by array constructor is given below.

```
<script>
var emp=new Array("Mahmood","Rabi","Saber");
document.write(emp[0] + "<br>");
document.write(emp[1] + "<br>");
document.write(emp[2] + "<br>");
</script>
```

JavaScript Array Methods

toString(): Convert array to string
join(): adds the value after each element of array
Pop(): remove the first element of the array
Push(): insert new element into the array
Length: return the number of elements in the array
Splice: add/remove elements from array
Concate(): combine more than one array
Sort(): arrange the elements of array

Array Methods

DEMO-3

JavaScript loop

- JavaScript supports different kinds of loops:
  - for - loops through a block of code a number of times
  - for/in - loops through the properties of an object
  - for/of - loops through the values of an alterable object
  - while - loops through a block of code while a specified condition is true
  - do/while - also loops through a block of code while a specified condition is true

Loop

DEMO-4

JavaScript Functions

- Function is a group of statement which is used to perform a specific task
- Advantage of JavaScript function
  - **Code reusability**: We can call a function several times so it saves coding.
  - **Less coding**: It makes our program compact.

JavaScript local and Global variables

- A JavaScript local variable is declared inside block or function. It is accessible within the function or block only. For example:

```
function abc(){
var x=10;//local variable
}
```

- A **JavaScript global variable** is accessible from any function. A variable i.e. declared outside the function is known as global variable. For example:

```
<script>
var data=200;//gloabal variable
function a(){
document.writeln(data);
}
a();//calling JavaScript function
</script>
```

JavaScript Function

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